Secondary 2

Computer Science

CA1 Term plan/ Assessment outline

This package contains

- a teaching program
- an assessment outline

Teaching program – Computer Science

Week	Unit Content	Learning Outcomes	Assessment tasks	Resources
Week 1	**1. Data representation** **1.1 Number systems** • Binary, Denary, and Hexadecimal systems • Binary addition & overflow • Logical binary shifts (left & right) • Two's complement signed binary **Class Activities** • 1 homework is assigned • Binary conversion & shift worksheets	• Understand why computers use binary to represent data • Convert between binary, denary, and hexadecimal number systems • Perform addition of two positive 8-bit binary integers and identify addition overflow • Perform logical binary shifts (left and right) and explain their effects on binary values • Use two's complement notation to represent positive and negative 8-bit binary numbers • Explain the practical uses of hexadecimal (MAC addresses, IPv6, HTML colour codes, error codes)		Cambridge IGCSE Computer Science Coursebook *pg 2–24*
Week 2	**1. Data representation** **1.2 Text, sound and images** • Character sets: ASCII and Unicode • Representation of sound & sampling • Sample rate & sample resolution • Representation of bitmap images • Pixels, colour depth & image resolution • File size calculation (sound & images) **Class Activities** • Sound & image file size calculations • 2 handouts are provided	• Understand how and why computers represent text using character sets • Compare ASCII and Unicode character sets in terms of bit length and character coverage • Explain how sound waves are sampled and converted to digital data • Describe the effects of sample rate and sample resolution on sound quality and file size • Explain how bitmap images are constructed using pixels, colour depth, and image resolution • Calculate the file size of bitmap images and sound files in KiB, MiB, and GiB	Week 4 (CT 1)	Cambridge IGCSE Computer Science Coursebook *pg 25–31*

Week	Unit Content	Learning Outcomes	Assessment tasks	Resources
Week 3	**2. Data transmission** **2.1 Types and methods of data transmission** • Data packets (header, payload, trailer) • Packet switching & role of routers • Serial vs Parallel transmission • Simplex, Half-duplex & Full-duplex • Universal Serial Bus (USB & USB-C) **Class Activities** • Packet routing diagram activity • 2 handouts are provided	• Understand the structure of data packets (header, payload, trailer) and why data is packetised • Describe the process of packet switching and the role of routers in routing packets • Differentiate between serial and parallel transmission with advantages and disadvantages • Differentiate between simplex, half-duplex, and full-duplex transmission modes • Explain the benefits and limitations of using Universal Serial Bus (USB and USB-C)	Week 4 (CT 1)	Cambridge IGCSE Computer Science Coursebook *pg 45–53*
Week 4	**3. Hardware** **3.1 Computer architecture (Part 1)** • Role of CPU, ALU, CU, System Clock • Registers (PC, MAR, MDR, CIR, ACC) • System Buses: Address, Data, Control • Fetch-Decode-Execute (FDE) cycle **Class Activities** • FDE cycle trace diagram activity • 1 handout is provided	• Explain the role of the CPU in processing instructions and data • Describe the functions of the ALU, Control Unit (CU), and System Clock • Explain the purpose and operation of CPU registers (PC, MAR, MDR, CIR, ACC) • Describe system buses (Address, Data, Control) including directionality and bus width • Explain each step of the Fetch-Decode-Execute (FDE) cycle and register transfers	Week 6 (CT 2)	Cambridge IGCSE Computer Science Coursebook *pg 75–81*
Week 5	**3. Hardware** **3.1 Computer architecture (Part 2)** • Factors affecting CPU performance • Clock speed, Cache size, Cores • Overclocking and its risks • Embedded systems & microcontrollers • Applications of embedded systems **Class Activities** • Embedded systems case studies • 2 handouts are provided	• Explain how clock speed, cache size, and the number of cores affect CPU performance • Understand overclocking and its potential risks (overheating, system instability) • Explain what an embedded system is and compare microcontrollers with general-purpose CPUs • Identify and analyze embedded systems in everyday devices (cars, appliances, vending machines) • Discuss the advantages and disadvantages of embedded systems	Week 6 (CT 2)	Cambridge IGCSE Computer Science Coursebook *pg 81–87*
Week 6	**Revision & Exam Prep** • Consolidate Ch 1 (1.1, 1.2) • Consolidate Ch 2 (2.1)	• Review and consolidate knowledge across Data Representation, Data Transmission, and Computer Architecture	CT 2 & CA1 Prep	Cambridge IGCSE Computer Science

Week	Unit Content	Learning Outcomes	Assessment tasks	Resources
	• Consolidate Ch 3 (3.1) • Mock exam questions & review	• Apply theoretical knowledge to solve structured Cambridge IGCSE examination questions • Identify areas for improvement and receive targeted feedback		Coursebook *Revision Worksheets*

Assessment Outline Template

Subject: Computer Science

Class: Secondary 2

Assessment type	Assessment Task	Weightings	Week
CT 1	Chapter 1: Data representation (1.1 Number systems, 1.2 Text, sound and images)	10%	4
CT 2	Chapter 2: Data transmission (2.1 Types and methods of data transmission) & Chapter 3: Hardware (3.1 Computer architecture)	10%	6
MCW (Formative)	Number systems, Data representation, and Types of data transmission	5%	7
Homework Submission	Binary, denary, and hexadecimal conversions; Character sets (ASCII/Unicode); Data packet structures; FDE cycle steps; and Embedded systems analysis	5%	Combined
CA1	**Comprehensive Assessment:** • 1.1 Number systems • 1.2 Text, sound and images • 2.1 Types and methods of data transmission • 3.1 Computer architecture	**70%**	**Week 9**